



SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 04

GROUP NO: 09

PROJECT TITLE: Development and Comparative Evaluation of a Cost-Aware Ensemble Machine Learning Framework for Anaemia Prediction Using DHS/NFHS Data

PROJECT ABSTRACT:

Anaemia remains a major public health concern in developing regions, particularly in low-resource settings where access to advanced diagnostic infrastructure is limited, making early and reliable identification challenging. This project aims to develop and comparatively evaluate a cost-aware ensemble machine learning framework for anaemia prediction using demographic and health-related variables derived from Demographic and Health Survey (DHS) and National Family Health Survey (NFHS) datasets. The study focuses on designing predictive models that maintain strong performance while remaining feasible for large-scale public health applications. Initially, baseline supervised learning models including Logistic Regression, Random Forest, Artificial Neural Networks, and gradient boosting techniques will be implemented to establish performance benchmarks. An existing hybrid stacking architecture proposed in prior research will be replicated under identical experimental conditions to ensure fair comparison. Subsequently, an enhanced ensemble stacking framework integrating additional boosting strategies will be developed to investigate potential improvements in predictive accuracy and generalization. Model performance will be evaluated using standardized metrics such as accuracy, Area Under the Curve (AUC), sensitivity, specificity, and cross-validation to ensure robustness and reliability. Furthermore, anaemia prediction performance will be analysed with specific reference to Odisha using NFHS data to assess regional applicability and public health relevance. The overall objective is to determine whether advanced ensemble learning approaches can improve anaemia prediction accuracy while supporting cost-conscious and data-driven healthcare analysis in resource-constrained environments.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

Software Specifications

- Programming Language: Python 3.13
- Development Environment: Visual Studio Code
- Libraries Used: NumPy, Pandas, Scikit-learn, XGBoost, LightGBM, CatBoost, Matplotlib / Seaborn, Statsmodels (if required)
- Operating System: Windows 11

Hardware Specifications

- Device: HP Victus Laptop
- Processor: AMD Ryzen Series Processor (Multi-core CPU architecture)
- RAM: 16 GB DDR4
- Storage: SSD-based system
- GPU: Not required (CPU-based model training and evaluation performed)

Methods / Algorithms Used

- Logistic Regression
- Random Forest Classifier
- Artificial Neural Network (MLP Classifier)
- XGBoost Classifier

(2) NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:

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Arnab Sahoo

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Aditya Animesh Sahoo

(3) APPROVAL STATUS (To be filled in by the Section Coordinator of FRP):

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Project Supervisor

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Section Coordinator, FRP

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FRP Coordinator